

VISUAL BASIC

Some Important Functions

- **InputBox()** : InputBox() displays a small dialog box that asks the user to enter a value. The value entered by the user is returned by the function as a String.

syntax: InputBox(Prompt,[Title],[Default],[XPos],[YPos],[HelpFile],[Context]) As String

Example 1: Accepting text

```
Dim a As String
Private Sub Command1_Click()
    a = InputBox("Enter your name")
    MsgBox a
End Sub
```

If the user enters Pushkar, the InputBox returns the string "Pushkar", which is then displayed by MsgBox.

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The parameters enclosed in [] are optional.

Parameter	Description
Prompt	Message or instruction displayed to the user. Required
Title	Text displayed in the title bar of the InputBox. Optional
Default	Initial value displayed in the input box. Optional
XPos	Horizontal position of the InputBox on the screen. Optional
YPos	Vertical position of the InputBox on the screen. Optional
HelpFile	Name of the Help file associated with the InputBox. Optional
Context	Help context number used to identify the appropriate Help topic. Optional

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E>

Project1 - Form1 (Code)

Form Load

```
Private Sub Form_Load()  
  InputBox "Enter your name", "Student Information", "", 8000, 6000, "StudentHelp.hlp", 101  
End Sub
```

Student Information

Enter your name

OK
Cancel
Help



Example 2: Accepting a number

```
Dim a As String
Private Sub Command1_Click()
    a = InputBox("Enter a number")
    MsgBox (a)
End Sub
```

➤ Val()

Val() converts the numeric portion at the beginning of a string into a number. It is commonly useful when numeric data has been received as text from an InputBox or TextBox.

Syntax: Val(string)

Example:

```
a = InputBox("First no.")
b = InputBox("Second no.")
MsgBox a & " + " & b & " = " & (Val(a) + Val(b))
```





➤ Len()

- Len() returns the number of characters in a string.
- Syntax – left(
• Example:
 a = "Patna"
 MsgBox Len(a)
 MsgBox Len("Patna")
- Both examples return 5.

➤ Left()

- Left() returns a specified number of characters from the left side of a string.

Syntax:

Left(string, length)

Example:

a = "PATNA"

MsgBox Left(a, 3)

Result: PAT





➤ Right()

- Right() returns a specified number of characters from the right side of a string.

Syntax : Right(string, length)

Example:

a = "PATNA"

MsgBox Right(a, 3)

'Result: TNA

➤ Mid()

- Mid() returns a specified number of characters from a string, beginning at a specified position. Positions are counted from 1.

- Mid(string, start[, length])

- Example:

a = "PATNA"

MsgBox Mid(a, 2, 3)

'Result: ATN





Some Important String Function:

Function	Purpose	Example
Len()	Returns the number of characters	Len("PATNA") → 5
Left()	Returns characters from the left	Left("PATNA", 3) → "PAT"
Right()	Returns characters from the right	Right("PATNA", 3) → "TNA"
Mid()	Returns characters from a specified position	Mid("PATNA", 2, 3) → "ATN"
Val()	Converts the leading numeric part of text to a number	Val("25") → 25
UCase()	To convert into upper case	MsgBox UCase("hello")
LCase()	To convert into lower case	MsgBox LCase("HEllo")
Asc()	Returns the ANSI/ASCII code of the first character in a string.	MsgBox Asc("A") '65 MsgBox Asc("a") '97
Chr()	Returns the character of ASCII code	MsgBox Chr(65) 'A
InStr()	Find the position of one string or character inside another string.	MsgBox InStr("RAMAN", "M")

Some Important Math Function: **VISUAL BASIC**



Function	Description	Example
Abs()	Returns the absolute value of a number.	Abs(-25) $\rightarrow$ 25
Atn()	Arc Tangent (Inverse Tangent).	Atn(1) $\rightarrow$ 0.785398
Cos()	Returns the cosine of an angle.	Cos(0) $\rightarrow$ 1
Exp()	Exponential function.	Exp(1) $\rightarrow$ 2.7182818284590
Fix()	Returns integer portion by truncating decimals toward zero.	Fix(-6.34) $\rightarrow$ -6
Int()	Returns greatest integer less than or equal.	Int(-6.34) $\rightarrow$ -7
Log()	Natural logarithm.	Log(10) $\rightarrow$ 2.302585
Rnd()	Returns a random number.	Rnd() $\rightarrow$ 0 to less than 1
Sgn()	1 if positive, 0 if zero, -1 if negative.	Sgn(-10) $\rightarrow$ -1
Sin()	Returns the sine of an angle.	Sin(0) $\rightarrow$ 0
Sqr()	Returns the square root of a positive number.	Sqr(81) $\rightarrow$ 9
Tan()	Returns the tangent of an angle.	Tan(0) $\rightarrow$ 0

DATE & TIME FUNCTIONS

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Function	Description	Example
Date()	Returns the current date.	MsgBox Date()
Time()	Returns the current time.	MsgBox Time()
Now()	Returns the current date and time.	MsgBox Now()
DateAdd()	Adds a specified time interval to a date.	MsgBox DateAdd("d", 7, #April 30, 2000#)
DateDiff()	Returns the difference between two dates for a specified interval.	MsgBox DateDiff("d", #April 30, 2000#, #May 5, 2000#)
DatePart()	Returns a specified part of a date, such as quarter, month, year or weekday.	MsgBox DatePart("q", #April 30, 2000#)
DateSerial()	Creates a date from year, month and day values.	MsgBox DateSerial(2000, 4, 30)
DateValue()	Converts a date string into a Date value.	MsgBox DateValue("April 30, 2000")
TimeSerial()	Creates a time from hour, minute and second values.	MsgBox TimeSerial(10, 30, 0)
TimeValue()	Converts a time string into a Time value.	MsgBox TimeValue("10:30:00 AM")
Timer()	Returns the number of seconds elapsed since midnight.	MsgBox Timer()



Format\$ Function

Statement	Output
Format\$(Now,"m-d-yy")	1-1-00
Format\$(Now,"m/d/yy")	1/1/00
Format\$(Now,"mm-dd-yy")	01-01-00
Format\$(Now,"d mmm, yyyy")	1 Jan, 2000
Format\$(Now,"hh:mm:ss AM/PM mm-dd-yy")	01:00:00 AM 01-01-00

Note : Now in VB6 is a built-in function that returns the current date and time of the computer. Directly printing current date and time will print in long format. Format\$() function gives control over exact format to be print.

Example:

MsgBox Now

might display:19-09-2026 12:18:35





Assignment : Write a program to enter a word using InputBox and display:

- (1) its length
- (2) its first two characters,
- (3) its last two characters, and
- (4) the characters from position 2 to 4.

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MsgBox() in Visual Basic 6.0

- MsgBox() is a built-in VB6 function used to display a message or ask a question to the user.
- It can also return the button selected by the user.
- It is commonly used for information, warnings, errors, confirmations, and other messages.0

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Syntax of MsgBox()

MsgBox(Prompt, [Buttons], [Title], [HelpFile], [Context]) As VbMsgBoxResult

- Arguments enclosed in [] are optional.
- For basic VB6 programs, Prompt, Buttons, and Title are used most often.





Arguments of MsgBox()

- Prompt – The message to be displayed. Required.
- Buttons – Specifies which buttons and icon will appear. Optional.
- Title – Specifies the title of the message box. Optional.
- HelpFile – Specifies the Help file associated with the message box. Optional.
- Context – Specifies the Help context number. Optional.

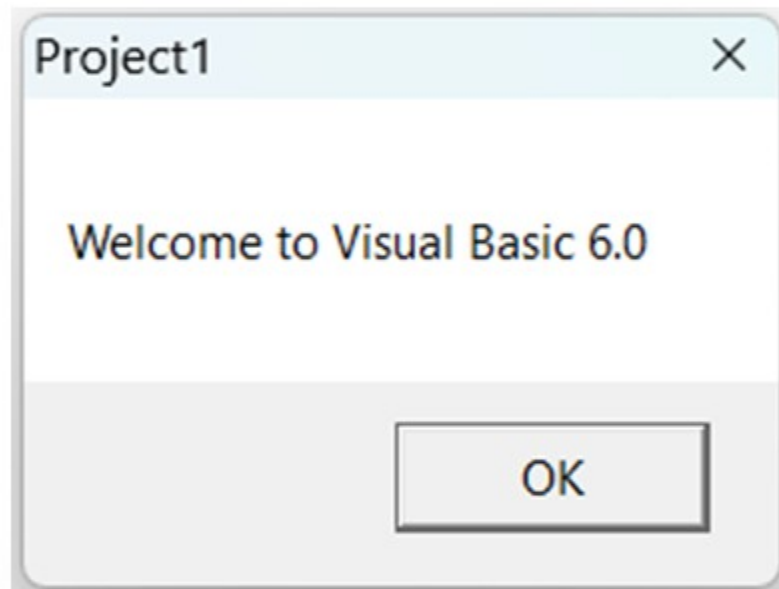




Simple MsgBox

```
Private Sub Command1_Click()  
    MsgBox "Welcome to Visual Basic 6.0"  
End Sub
```

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Note:

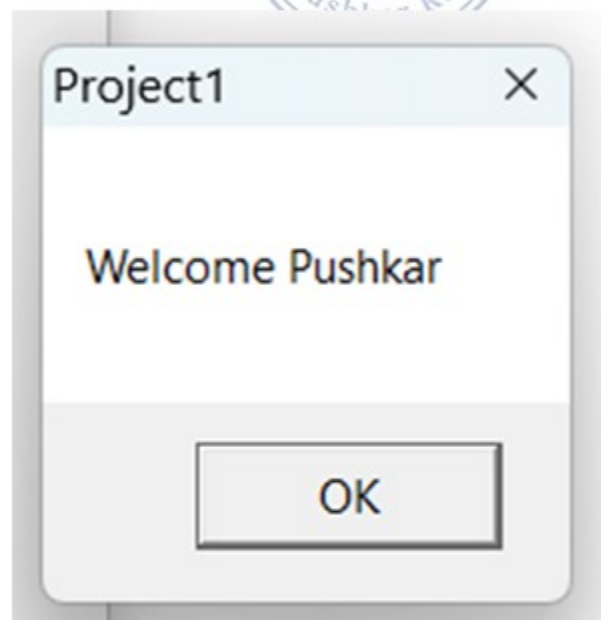
- Only the message (Prompt) is supplied.
- By default, VB6 displays an OK button.



Displaying a Message from a Variable

```
Private Sub Command1_Click()  
    Dim name As String  
    name = "Pushkar"  
    MsgBox "Welcome " & name  
End Sub
```

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The & operator joins the text "Welcome " with the value stored in name.

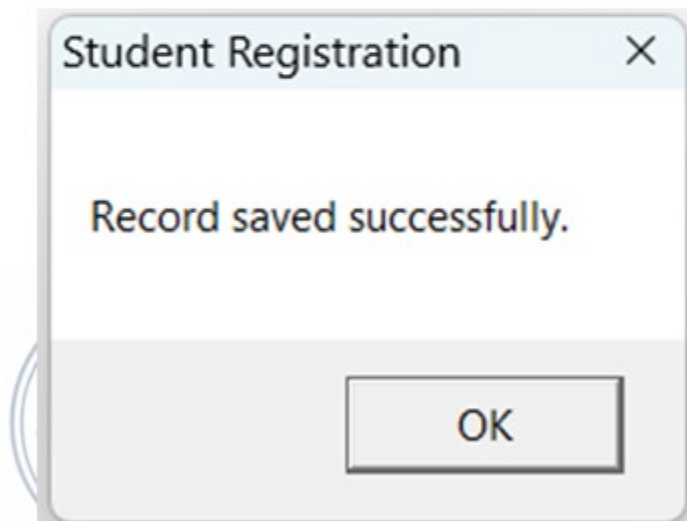
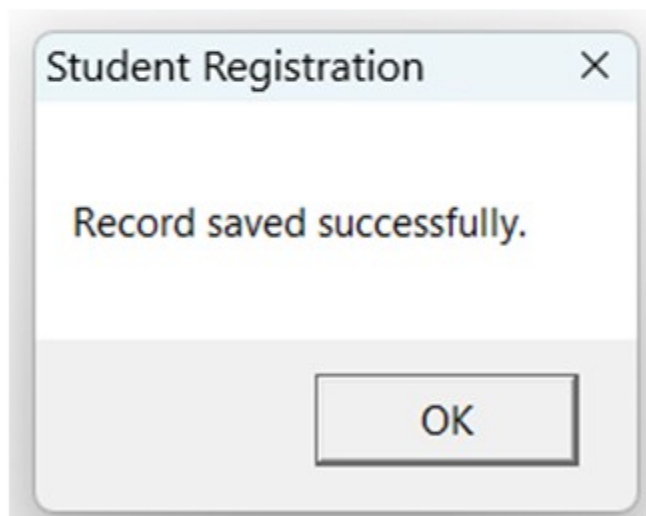


Using the Title Argument

MsgBox "Record saved successfully.", , "Student Registration"

MsgBox "Message", , "Title"

MsgBox "Record saved successfully.", vbOKOnly, "Student Registration"



The second argument is skipped when the default button settings are required.



Buttons in MsgBox()

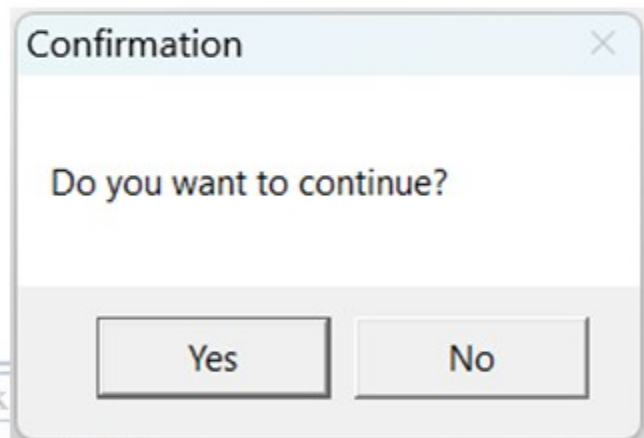
- vbOKOnly – OK
- vbOKCancel – OK, Cancel
- vbYesNo – Yes, No
- vbYesNoCancel – Yes, No, Cancel
- vbRetryCancel – Retry, Cancel
- vbAbortRetryIgnore – Abort, Retry, Ignore





Example: Yes/No Buttons

MsgBox "Do you want to continue?", vbYesNo, "Confirmation"



- The message box displays Yes and No buttons.



Icons in MsgBox()

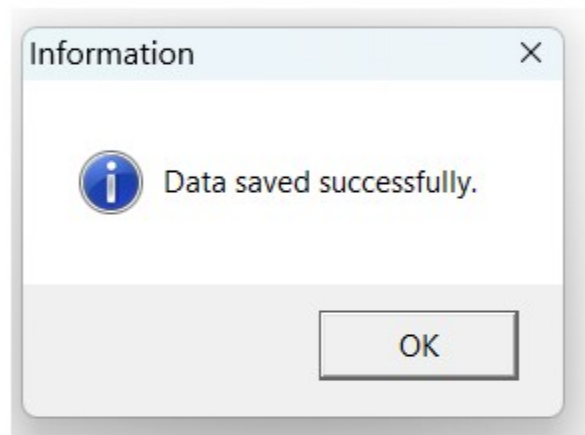
- The Buttons argument can also specify an icon.
- vbInformation – Information
- vbQuestion – Question
- vbExclamation – Warning
- vbCritical – Critical/Error





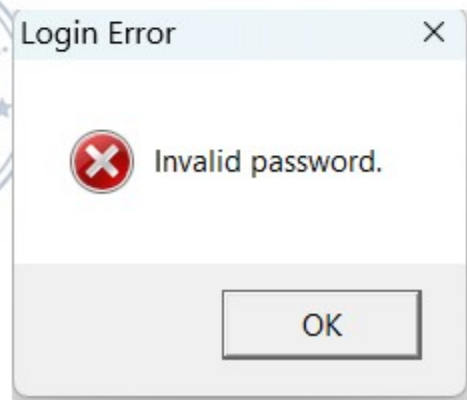
Examples of Icons

MsgBox "Data saved successfully.", vbInformation, "Information"



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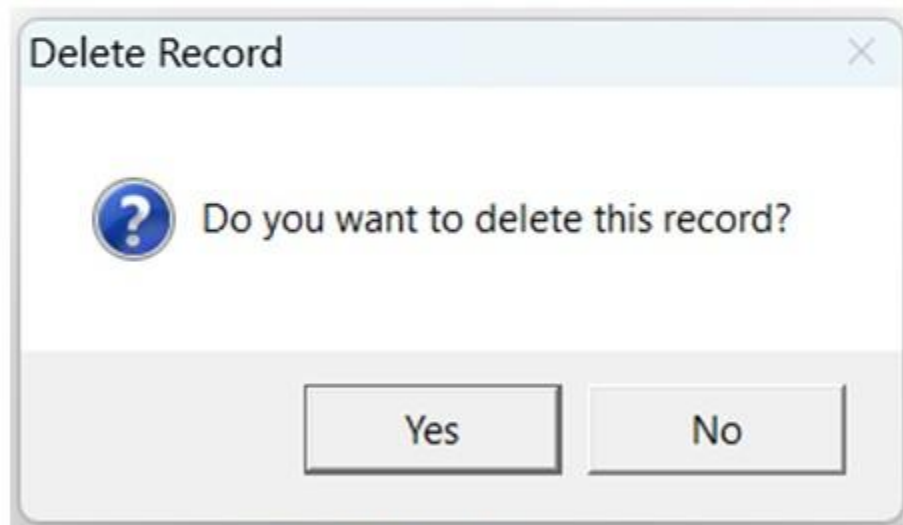
MsgBox "Invalid password.", vbCritical, "Login Error"





Combining Button and Icon

MsgBox "Do you want to delete this record?",vbYesNo + vbQuestion,"Delete Record"



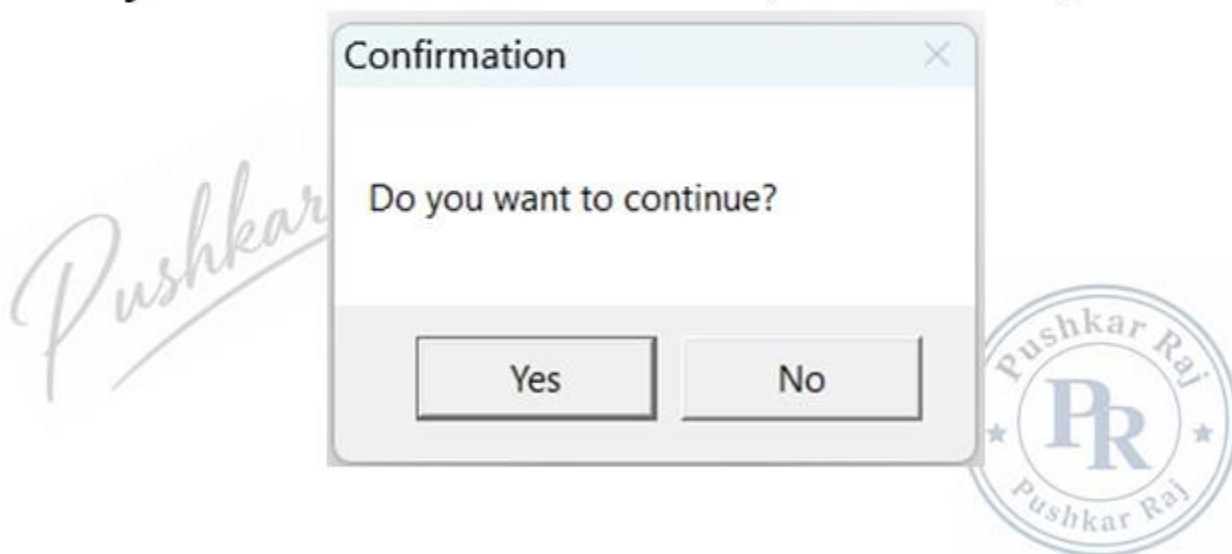
- vbYesNo → displays Yes and No.
- vbQuestion → displays a question icon.
- The title is "Delete Record".
- Button and icon settings can be combined in the Buttons argument.



Getting the User's Response

Dim ans As Integer

ans = MsgBox("Do you want to continue?", vbYesNo, "Confirmation")



If the user clicks Yes → ans receives vbYes

If the user clicks No → ans receives vbNo



Using the Returned Value

Dim ans As Integer

ans = MsgBox("Do you want to continue?", vbYesNo, "Confirmation")

If ans = vbYes Then

MsgBox "You selected Yes."

Else

MsgBox "You selected No."

End If





Important Distinction

- vbYesNo specifies which buttons should be displayed.
- vbYes represents the Yes button selected by the user.
- vbNo represents the No button selected by the user.

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Common Return Values

- vbOK – OK button clicked
- vbCancel – Cancel button clicked
- vbYes – Yes button clicked
- vbNo – No button clicked
- vbRetry – Retry button clicked
- vbIgnore – Ignore button clicked
- vbAbort – Abort button clicked





Practical Example – Delete Confirmation

```
Private Sub Command1_Click()
```

```
    Dim ans As Integer
```

```
    ans = MsgBox("Do you want to delete this record?", vbYesNo + vbQuestion, "Delete Record")
```

```
    If ans = vbYes Then
```

```
        MsgBox "Record deleted successfully.", _
```

```
            vbInformation, "Delete Record"
```

```
    Else
```

```
        MsgBox "Delete operation cancelled.", _
```

```
            vbInformation, "Delete Record"
```

```
    End If
```

```
End Sub
```





Working of the Delete Confirmation

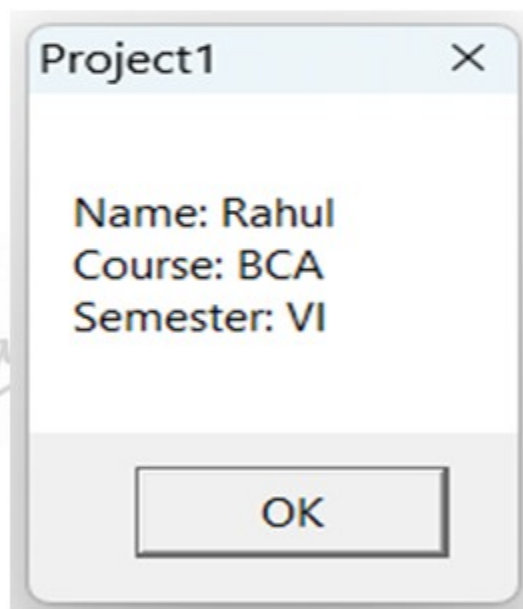
- The program displays a confirmation message.
- The user chooses Yes or No.
- MsgBox() returns the selected button.
- The If...Else statement checks the returned value.
- The program performs the appropriate action.





Displaying Multiple Lines

MsgBox "Name: Rahul" & vbCrLf & "Course: BCA" & vbCrLf & "Semester: VI"



- vbCrLf moves the following text to a new line.



MsgBox() with Calculated Values

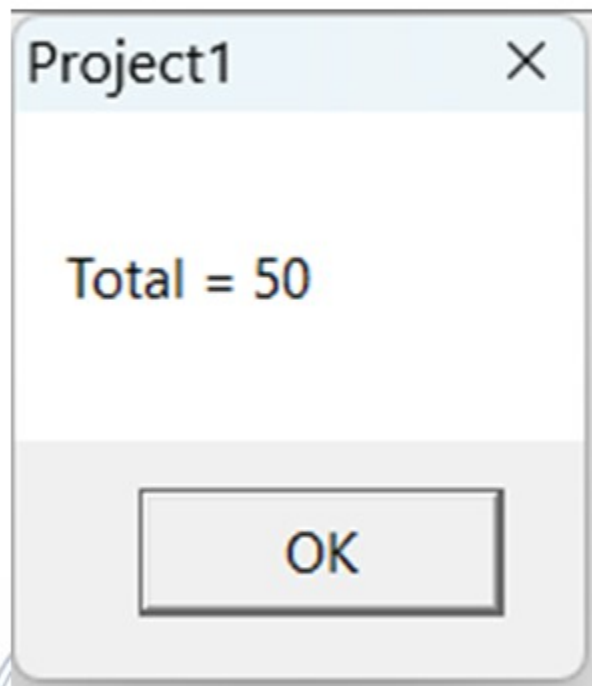
```
Dim a As Integer
```

```
Dim b As Integer
```

```
a = 20
```

```
b = 30
```

```
MsgBox "Total = " & (a + b)
```



- Output: Total = 50
- The calculation is performed first, and the result is joined with the text.



MsgBox() and CStr()

- VB6 can automatically convert a numeric value to text when it is joined with text using &.
- Example: MsgBox "Total = " & total
- Normally, MsgBox "Total = " & CStr(total) is unnecessary.
- CStr() can be used when an explicit conversion to String is required.
- Avoid unnecessary conversions.



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MsgBox()	InputBox()
Mainly displays a message or question.	Takes text input from the user
can display different buttons.	provides a text box for input
can return the selected button.	returns the entered text

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MsgBox() vs InputBox() – Examples

MsgBox "Record saved successfully."

name = InputBox("Enter your name")

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- MsgBox() asks the user to read or respond to a message.
- InputBox() asks the user to enter data.





Important Points to Remember

- MsgBox() is a built-in VB6 function.
- Prompt is the main and required argument.
- Buttons and Title are commonly used optional arguments.
- MsgBox() can display different button combinations and icons.
- MsgBox() can return the button selected by the user.
- vbYesNo specifies the buttons; vbYes and vbNo represent the response.
- vbCrLf can be used to display multiple lines.
- MsgBox() is generally modal.
- InputBox() is used when the program needs text input.





MsgBox() in Visual Basic 6.0

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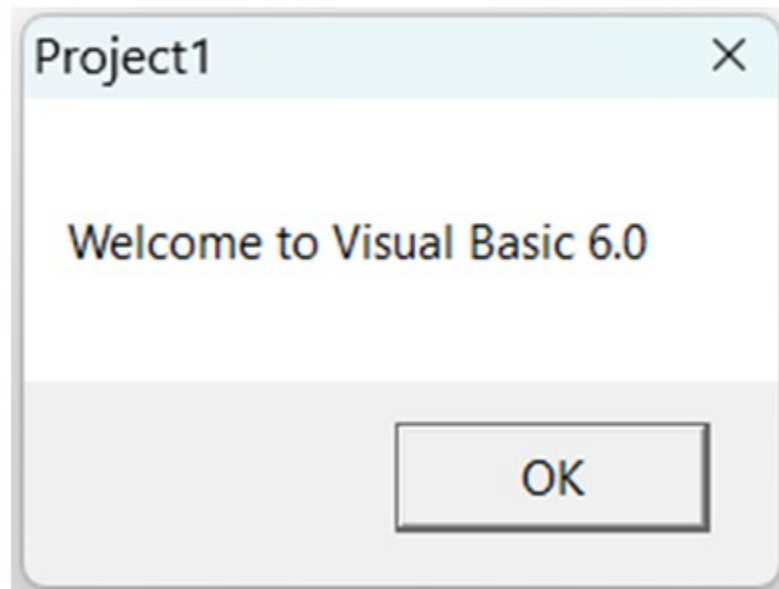




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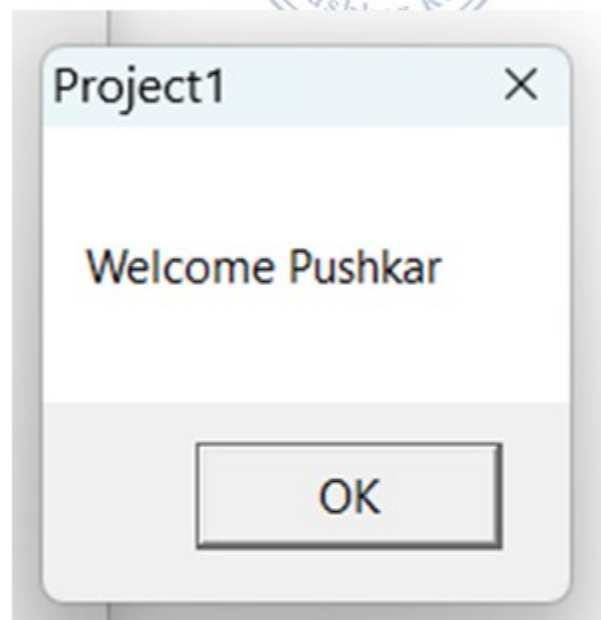
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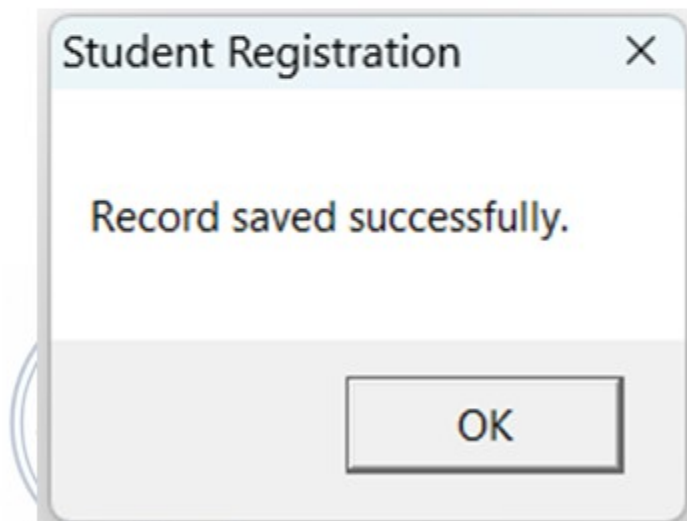
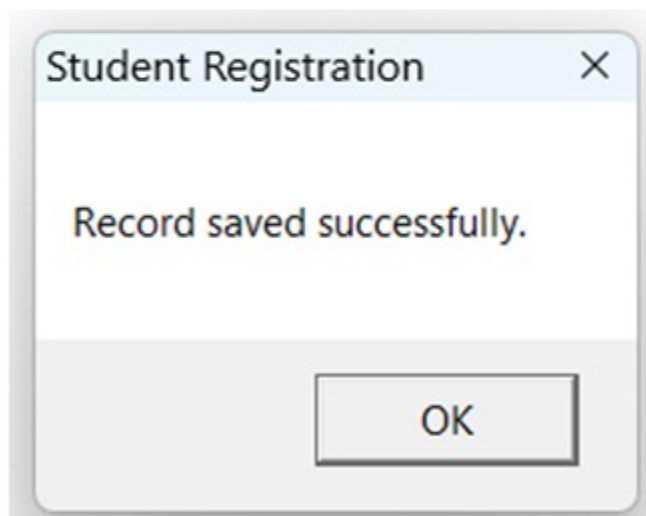


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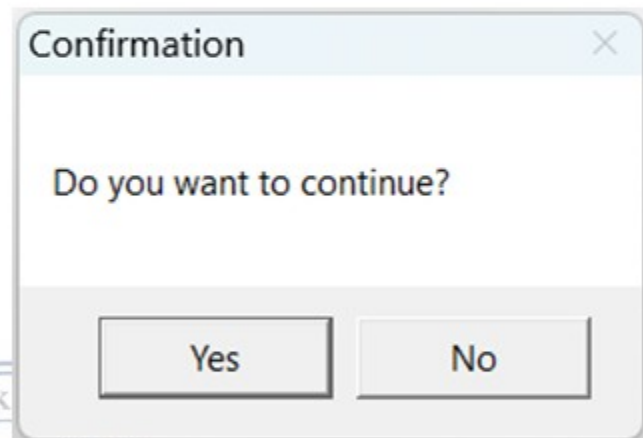
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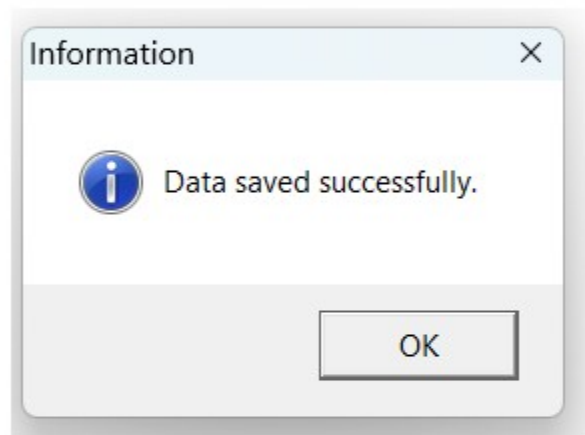
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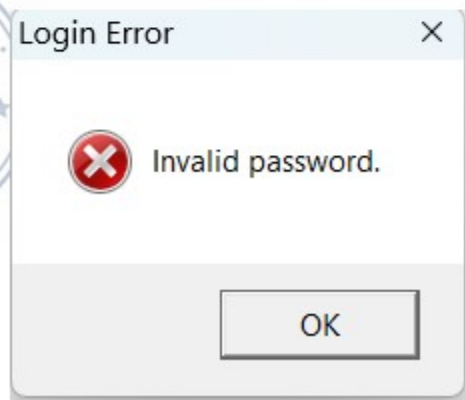
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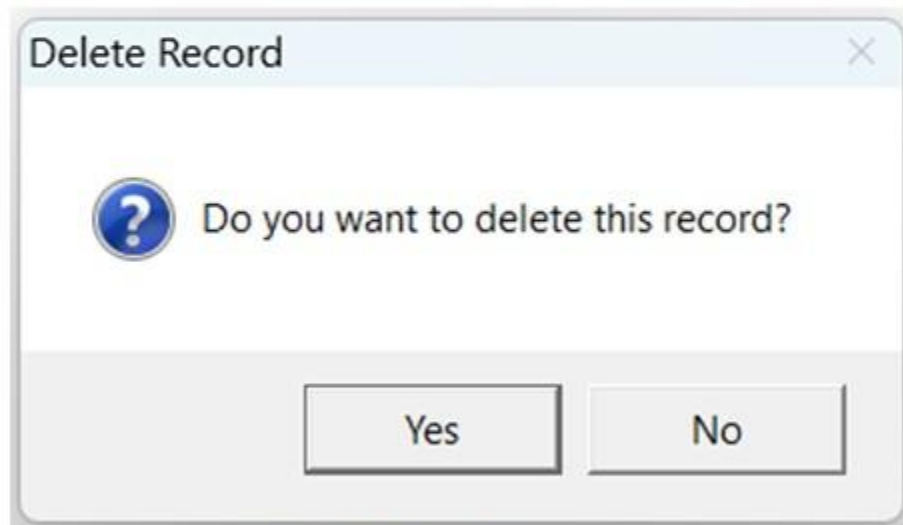
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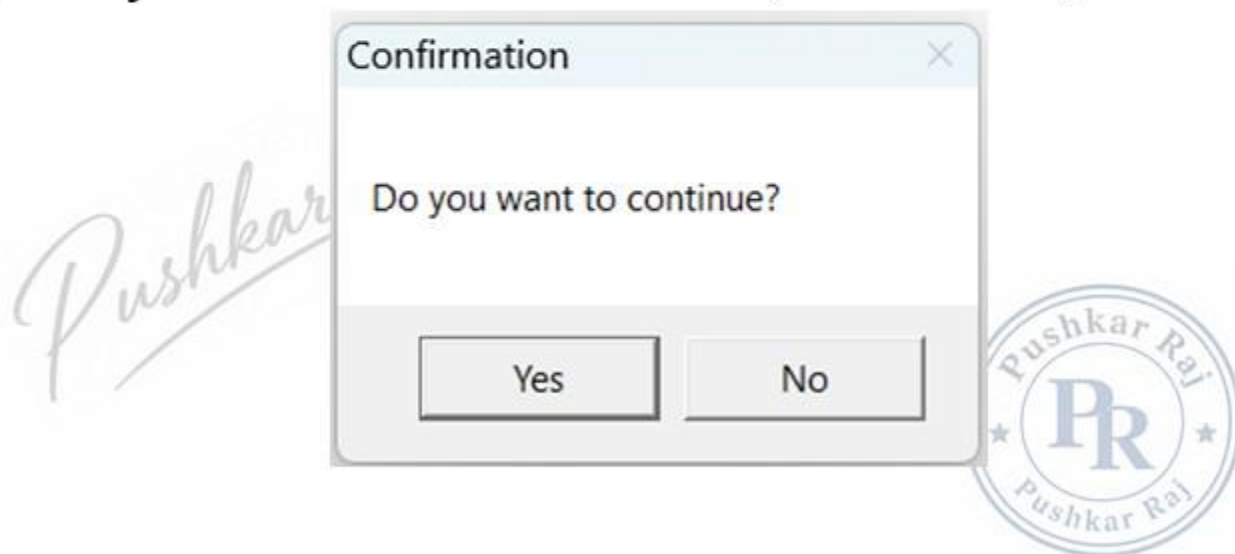
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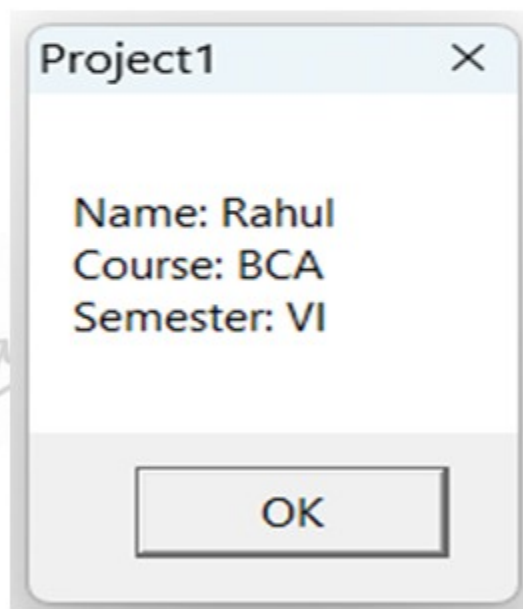
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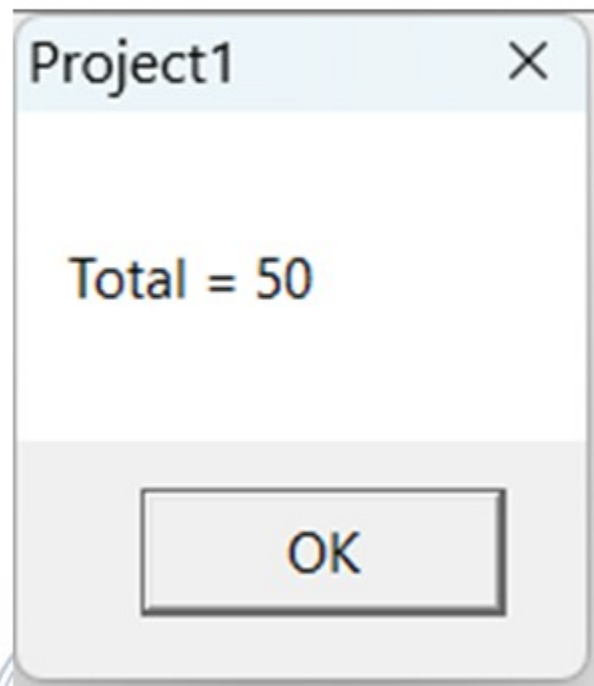
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